

13.7 Summary and Notes

13.7.1 Summary

Idea.

1. The error rates of a family, familywise, per-family and false discovery, are ordered (Definition 13.1.1, Proposition 13.1.1). Control should be strong; a gate such as the protected LSD gives only weak control (Definition 13.1.2, Proposition 13.1.3). Simultaneous intervals give strong control, including of sign errors (Proposition 13.1.2).
2. Scheffé: a family of estimable functions closed under linear combination is a subspace of $\mathcal{C}(\mathbf{X})$, and its largest squared t statistic is q times an $F(q, n - r)$ variable. The intervals with multiplier $\sqrt{qF_\alpha(q, \nu)}$ are exact, even for functions chosen after seeing the data (Theorem 13.2.2); the F test rejects iff some Scheffé interval excludes its hypothesized value (Corollary 13.2.3); smaller subspaces give shorter intervals (Section 13.2).
3. Tukey: in a balanced one-way layout the intervals $\bar{y}_k - \bar{y}_l \pm q_\alpha(g, \nu)s/\sqrt{m}$ are exact for all pairs, extend to all contrasts, and are the shortest of their form (Definition 13.3.1, Theorem 13.3.2, Proposition 13.3.3). The Tukey–Kramer intervals Eq. 13.3.3 are conservative.
4. Bonferroni needs no assumption on dependence and bounds the expected number of false rejections (Theorem 13.4.1); Šidák’s level is valid for independent estimates, and a shared s only helps (Proposition 13.4.3). Closed testing gives strong control (Theorem 13.4.4), and Holm’s step-down procedure, its Bonferroni case, rejects everything Bonferroni does and possibly more (Theorem 13.4.5).
5. The Benjamini–Hochberg procedure controls the false discovery rate at $\pi_0 q$ for independent null p -values (Theorem 13.5.2), and Benjamini and Yekutieli’s results extend it to positively dependent (PRDS) and arbitrarily dependent p -values.
6. Fix the family first: planned lists call for Bonferroni, Holm or exact multivariate t values, all pairs for Tukey, data snooping for Scheffé, screening for the false discovery rate (Section 13.6).

13.7.2 Notes and sources

Sources for this chapter. Christensen (2020, chapter 5) treats multiple comparisons within the projection framework used here, describing Scheffé’s method through a test space and showing that the F test rejects exactly when some single-degree-of-freedom hypothesis in that space does. Section 13.2 follows that view. His chapter also compares the least significant difference, Bonferroni, Tukey, Newman–Keuls and Duncan procedures and discusses the Fisherian and Neyman–Pearson views of multiple testing. Agresti (2015, section 3.5) gives a concise account of Bonferroni’s and Tukey’s methods, the Tukey–Kramer extension and the Benjamini–Hochberg procedure. The examples, simulations and the treatment of error rates, Šidák’s inequality and closed testing are this book’s own.

Scheffé and Tukey. Scheffé (1953) introduced the method for all contrasts, and his book (Scheffé 1959) develops it for general linear hypotheses and relates it to the F test and the

confidence ellipsoid. Tukey's 1953 manuscript circulated for decades before it appeared in his collected works (Tukey 1994). Kramer (1956) proposed the extension to unequal group sizes, and Hayter (1984) proved Tukey's conjecture that these intervals are conservative. The case of estimates with general correlation, where neither form is exact, is handled by computing multivariate t probabilities; Hothorn, Bretz and Westfall (2008) describe a general framework and Bretz, Hothorn and Westfall (2010) a practical guide. Dunnett (1955) gave the exact procedure for comparisons with a control.

Bonferroni, Šidák and Holm. The name commemorates Bonferroni's (1936) work on inequalities for probabilities of unions. Dunn (1961) popularized its use for simultaneous confidence intervals. Šidák (1967) proved the inequality for rectangular regions of the multivariate normal distribution that underlies Proposition 13.4.3 for correlated estimates. Holm (1979) introduced the step-down procedure, and Marcus, Peritz and Gabriel (1976) the closure principle, of which it is a case. Hochberg (1988) proposed the step-up version, whose validity rests on the inequality of Simes (1986).

False discovery rate. Benjamini and Hochberg (1995) defined the false discovery rate and proved Theorem 13.5.2. The leave-one-out proof given here is a standard later simplification. Benjamini and Yekutieli (2001) proved control under positive regression dependence and gave the version valid under arbitrary dependence. Storey (2002) introduced estimation of π_0 and the q -value, and Benjamini, Krieger and Yekutieli (2006) gave adaptive procedures with proven control.

Books. Miller (1981) is the classic account of simultaneous inference, and Hochberg and Tamhane (1987) and Hsu (1996) are comprehensive monographs. All three treat the procedures of Duncan (1955) and of Newman (1939) and Keuls (1952), and the closed step-down procedures that improve on Tukey's. Gelman, Hill and Yajima (2012) give a Bayesian view of why hierarchical models reduce the need for multiplicity adjustments.

13.7.3 References

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